STM Exercise 2

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Exercise 1: You have a cylindrical component made of an Al alloy (**10 mm diameter, YS= 150 MPa**): which is the **maximum load** you can apply on this component without having plastic deformation? Consider a **safety factor of n = 2**.

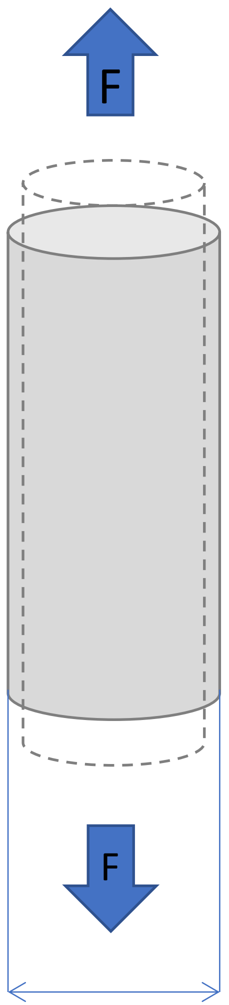
$$\sigma = \frac{F}{A}$$

where

- σ stress [Pa]
- F applied force [N]
- A cross-sectional area [m^2]

1-You have a cylindrical component made of an Al alloy (10 mm diameter, YS= 150 MPa): which is the maximum load you can apply on this component without having plastic deformation? Consider a safety factor of n = 2.

F=?



d0 = 10 mm

Yeld strength YS (σ_y)=150 Mpa
Safety factor =2

No plastic deformation $\rightarrow$ **σ<=σ_y**
Safety factor=2 $\rightarrow$ σ<=σ_y/2

Maximum stress:
σ_{max}= σ_y/2=150/2=75 Mpa

A=π**r²**=78.5 mm²

Maximum load:
F_{max}= σ_{max}*A= 75*78.5=5890 N

$$\sigma = \frac{F}{A}$$

where

- σ stress [Pa]
- F applied force [N]
- A cross-sectional area [m^2]

Poisson's ratio, ν

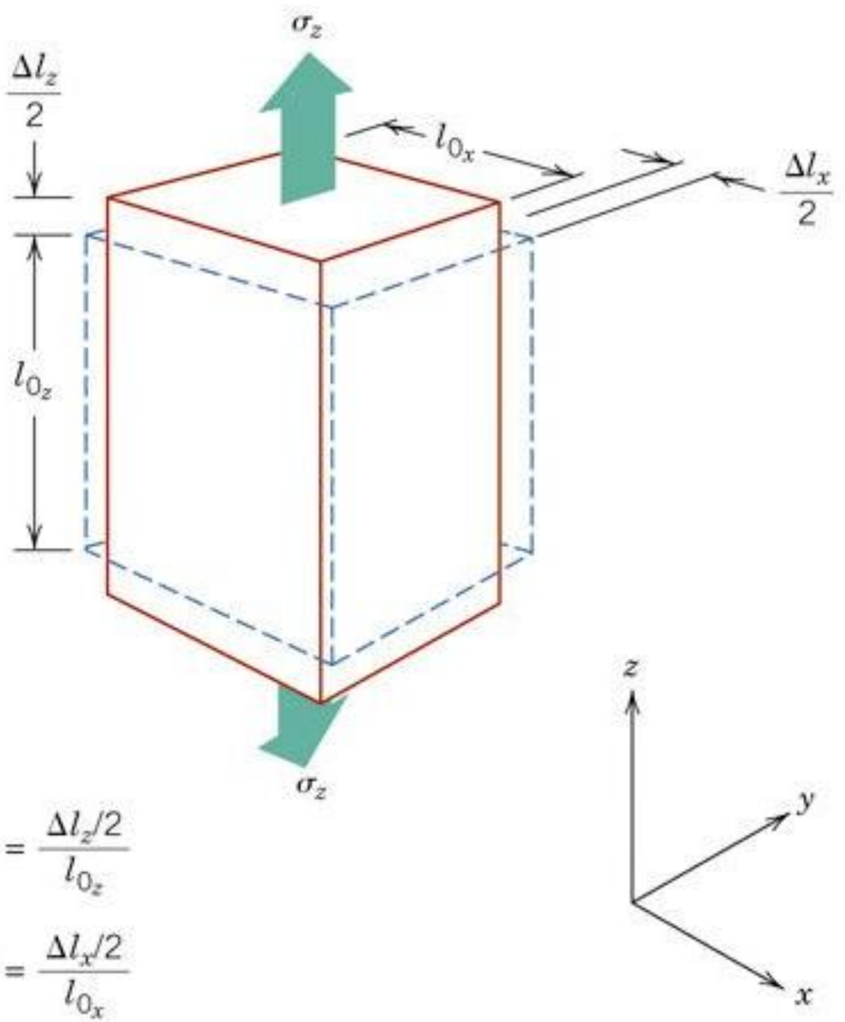
- **Poisson's ratio, ν:**

$$\nu = \frac{-\varepsilon_x}{\varepsilon_z} = \frac{-\varepsilon_y}{\varepsilon_z}$$

(Isotropic materials only)

metals: ν ~ 0.33
ceramics: ν ~ 0.25
polymers: ν ~ 0.40

Units:
E: [GPa] or [psi]
ν: dimensionless



$$\frac{\epsilon_z}{2} = \frac{\Delta l_z/2}{l_{0z}}$$

$$-\frac{\epsilon_x}{2} = \frac{\Delta l_x/2}{l_{0x}}$$

Exercise 2: You have a cylindrical aluminum component (**diameter =10 mm**, **v=0.34**, **E=70 GPa**): calculate the tensile load you need to apply to have a **2 µm** smaller diameter component.

$$\boxed{E = \frac{\sigma}{\varepsilon}}$$

$$\varepsilon = \frac{\Delta L}{L_0}$$

where

- ε strain
- ΔL total elongation [m]
- L_0 original length [m]

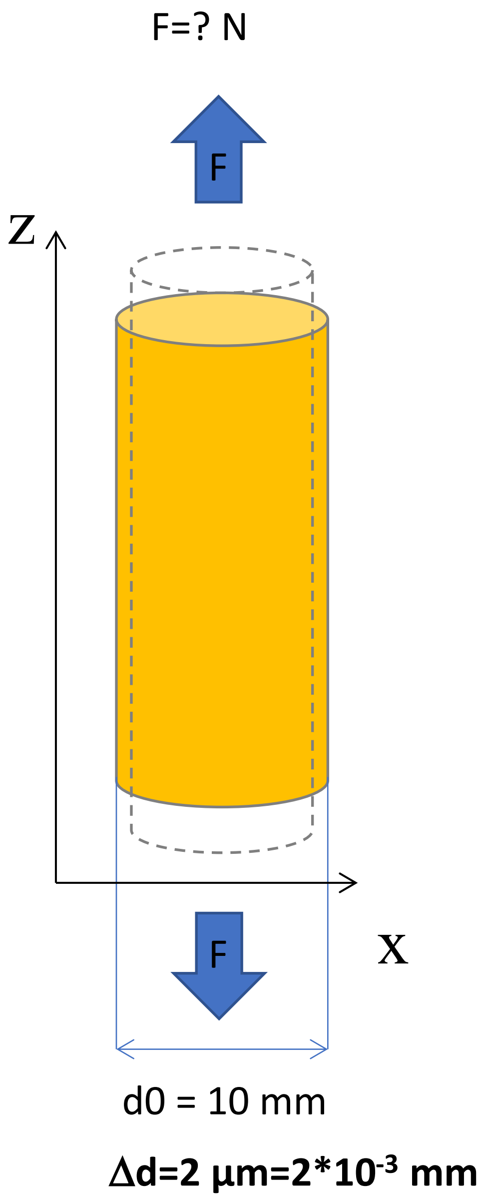
$$\sigma = \frac{F}{A}$$

where

- σ stress [Pa]
- F applied force [N]
- A cross-sectional area [m^2]

Poisson (ν) = - ε_x/ ε_z

2- You have a cylindrical aluminum component (**diameter =10 mm, $\boldsymbol{\upsilon}$=0.34, E=70 GPa**): calculate the tensile load you need to apply to have a **2 μm** smaller diameter component.



Poisson (ν) = - ε_x/ ε_z

When a load F is applied the component elongates along z and reduce its diameter along x.

Δd=2 μm=2*10^{-3} mm

$$\varepsilon_x = \frac{\Delta d}{d_0} = \frac{-2\times10^{-3}mm}{10mm} = -2\times10^{-4}$$

(*negative deformation for diameter reduction*)

ν = - ε_x/ ε_z

$$\varepsilon_z = -\frac{\varepsilon_x}{\upsilon} = -\frac{(-2\times10^{-4})}{0.34} = 5.8\times10^{-4}$$

X

$\sigma=\varepsilon_z E= (5.8 \times 10^{-4}) (70 \times 10^3 \text{ MPa}) = 40.6 \text{ MPa}$

A=πr^2=78.5 mm^2

F= σ*A= 40.6*78.5= **3187** N

Exercise 3: Which is the diameter you need for a **380 mm** long Cu component (**E=110 GPa)** to be loaded at **6600N**, if you can accept an **elongation of 0.5 mm**.

$$\boxed{E = \frac{\sigma}{\varepsilon}}$$

$$\varepsilon = \frac{\Delta L}{L_0}$$

where

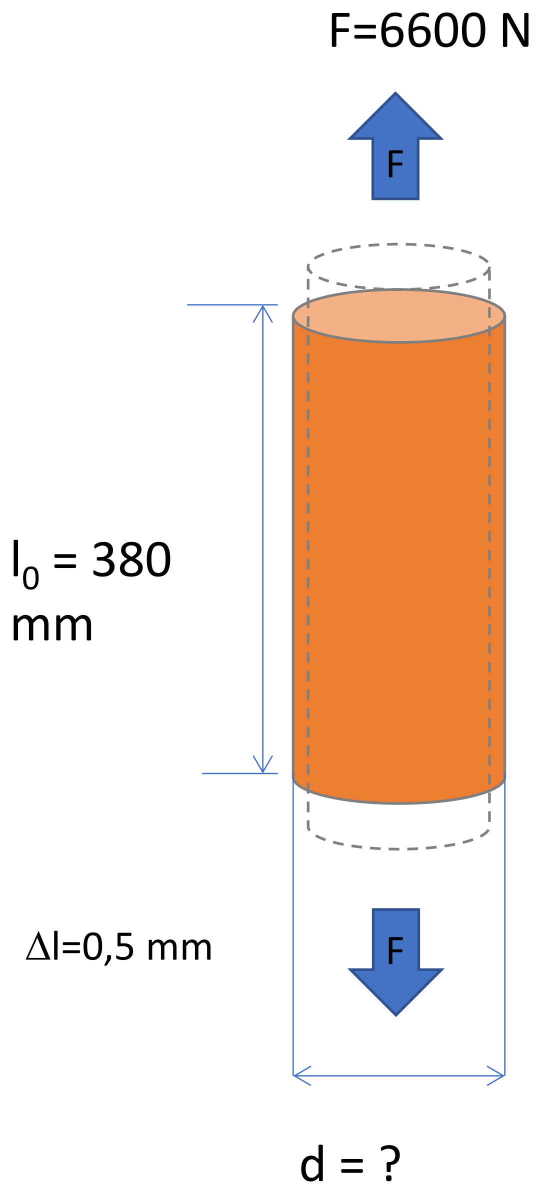
- ε strain
- ΔL total elongation [m]
- L_0 original length [m]

$$\sigma = \frac{F}{A}$$

where

- σ stress [Pa]
- F applied force [N]
- A cross-sectional area [m^2]

3- Which is the diameter you need for a 380 mm long Cu component (E=110 GPa) to be loaded at 6600N, if you can accept an elongation of 0.5 mm.



$\varepsilon=\Delta l/l_0=0.5/380=0.0013$

E=110 Gpa→110 000MPa

$\sigma=E*\varepsilon=110\,000*0.0013=144.7$ MPa

$\sigma=F/A$

$A=F/\sigma=6600/144.7=45.6$ mm^2

$A=\pi*r^2$

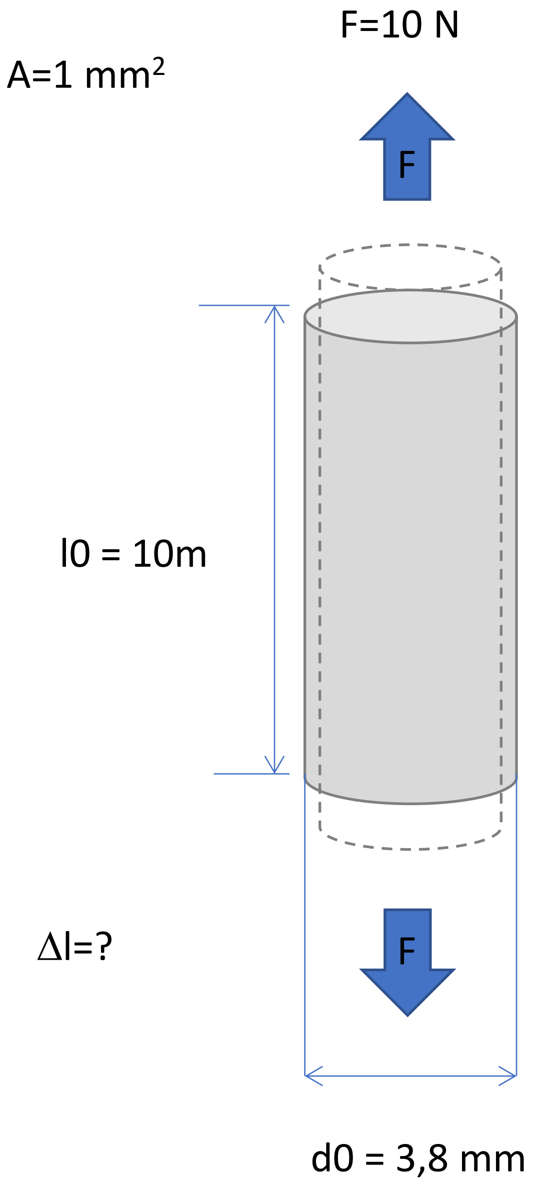
$r=\sqrt{\frac{A}{\pi}}=3.8$mm

$d=2*r=7.6$mm

Exercise 4: The elastic modulus of a **steel is 193 GPa**, of a **bronze is 100 GPa**, and of a **polymer is 0,17 GPa** ;

calculate the elastic elongation of a **10 meter long** wire made of steel, bronze or polymer , cross section is **1 mm^2**, the applied load is **F= 10 N**.

The elastic modulus of a steel is 193 GPa, of a bronze is 100 GPa, and of a polymer is 0,17 GPa ; **calculate the elastic elongation** of a 10 meter long wire made of steel, bronze or polymer , cross section is 1 mm^2, the applied load is F= 10 N.



σ=F/A=10/1=10 MPa

it is the same for all the materials

$\varepsilon=\sigma/E$

It depends on the Elastic Modulus (E), different for each material:

ε_{steel}=10/193 000 = 5.18*10^{-5}

$\varepsilon=\Delta l/l_0$

$\Delta l= \varepsilon * l_0$

$\rightarrow \Delta l_{steel}$= 5.18*10^{-5}*10000=0.518mm

ε_{bronze}=10/100 000=0.0001

$\rightarrow \Delta l_{bronze}$= 0.0001*10000= 1mm

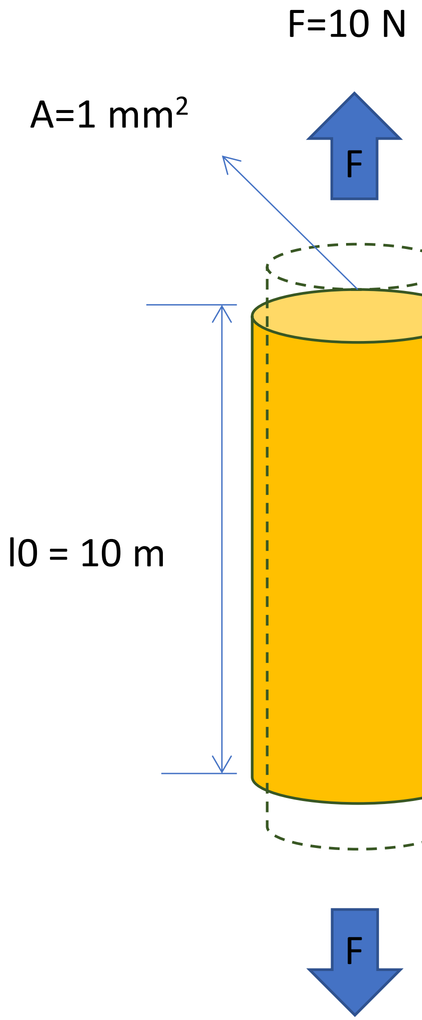
$\varepsilon_{polymer}$=10/170=0.06

$\rightarrow \Delta l_{polymer}$= 0.06*10000=600mm

Exercise 5: A wire 10 m long with a cross section of $1mm^2$ underwent tensile loading of **10N.**

Determine stress, strain and elongation if the wire is made of **Copper (E=110 Gpa)** or **steel (E=210 Gpa).**

Compare different material on the available excel file.



σ= F/A= 10 [N]/1 [mm^2] = **10 MPa**

If the wire is made of **copper** (E= 110 GPa) the elastic deformation is:

σ = E*ε →

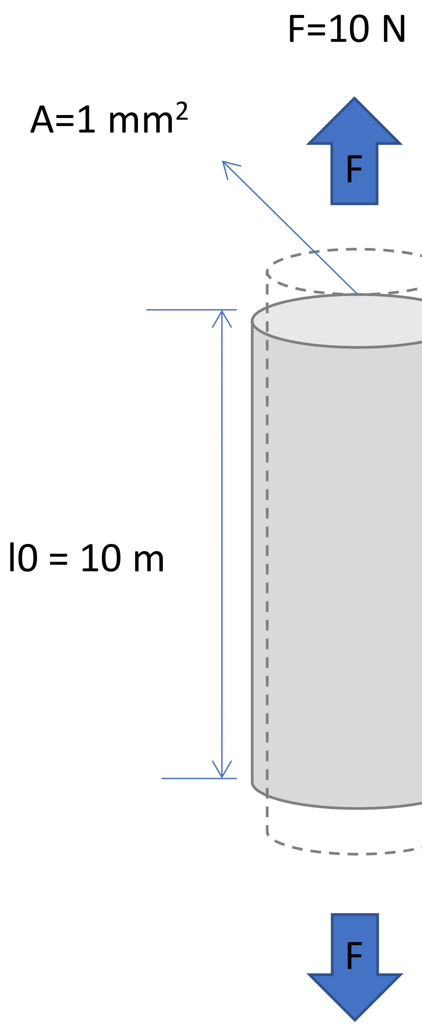
ε =σ / E

ε= 10 [MPa]/110 000 [MPa] = **0.00009 = 0.009 %**

ε =Δl/lo $\Delta l= \varepsilon * l_0$

The total elongation on 10 m is

0.00009 x 10 m=0.0009 m= **0.9 mm**



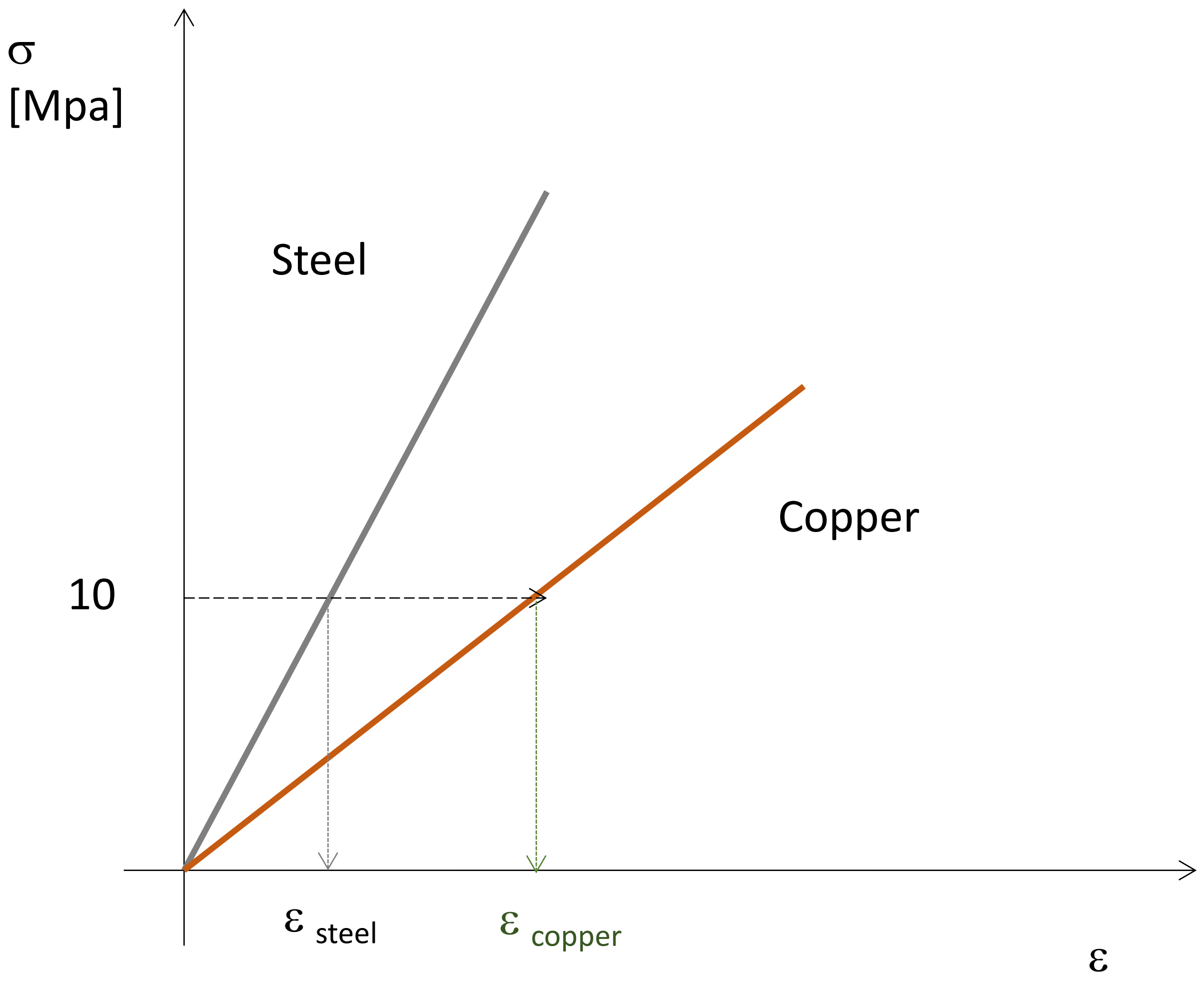
If the wire is made of **steel** (E=210 GPa):

ε= 10 /210 000=0.000048

The total elongation on 10 m is

0.000048 x 10 m = 0.00048 m=**0.48 mm**

<u>Since the elastic modulus of steel is almost double of the one of copper, the elastic deformation, in the same conditions, is almost a half.</u>

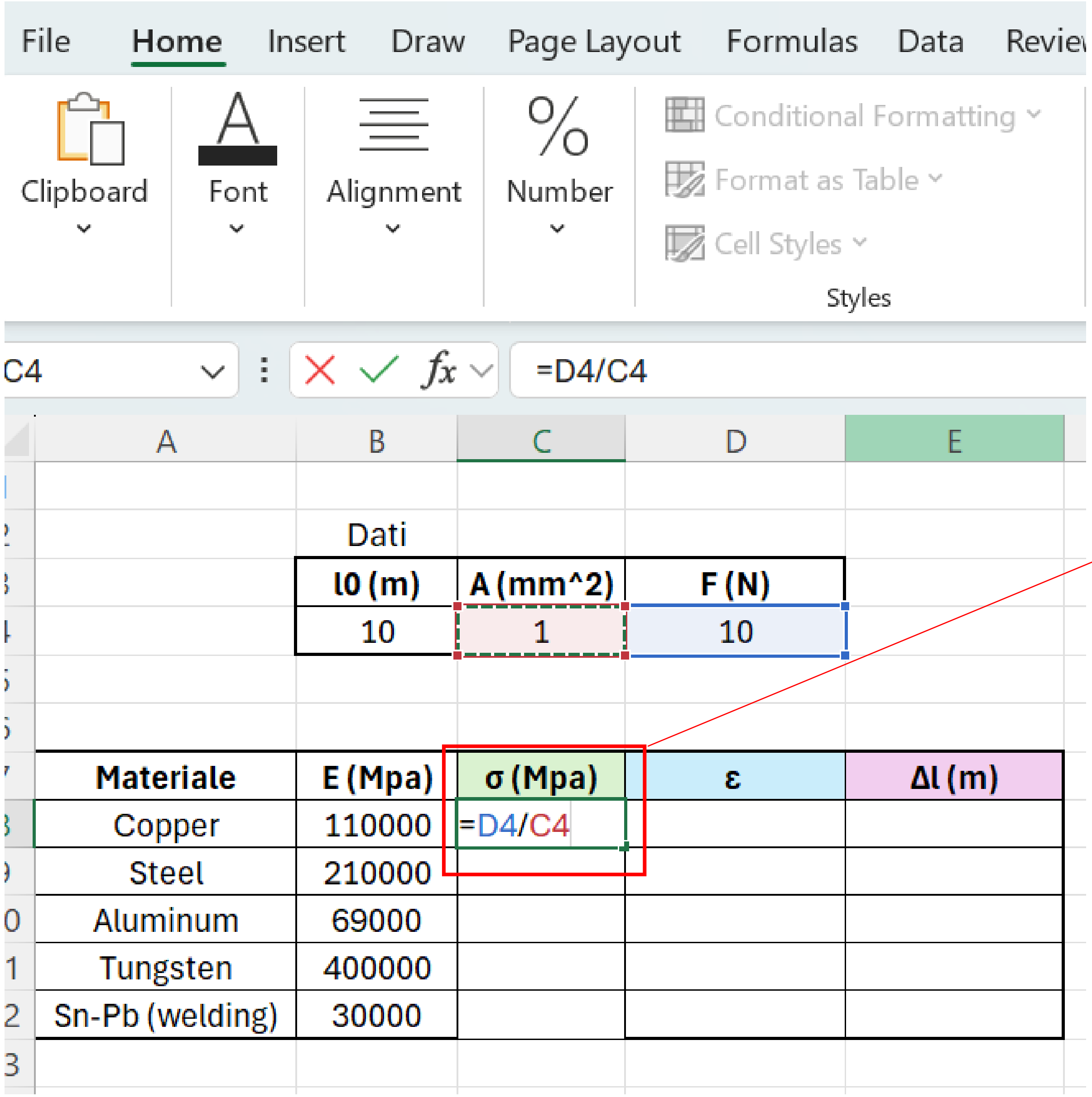

σ
[Mpa]
Steel
Copper
10
ε steel
ε copper
ε

Exercise 5: A wire 10 m long with a cross section of $1mm^2$ underwent tensile loading of **10N.**

Determine stress, strain and elongation if the wire is made of **Copper (E=110 Gpa)** or **steel (E=210 Gpa).**

Compare different material on the available excel file.

Go to «es5_exercise2_working»



$$\sigma = \frac{F}{A}$$

where

- σ stress [Pa]
- F applied force [N]
- A cross-sectional area [m^2]

Dati

l0 (m)	A (mm^2)	F (N)
10	1	10

Materiale	E (Mpa)	σ (Mpa)	ε	Δl (m)
Copper	110000	10	=C8/B8	
Steel	210000	10		
Aluminum	69000	10		
Tungsten	400000	10		
Sn-Pb (welding)	30000	10		

$$\varepsilon = \frac{\sigma}{E}$$

$$\varepsilon = \frac{\Delta L}{L_0}$$

where

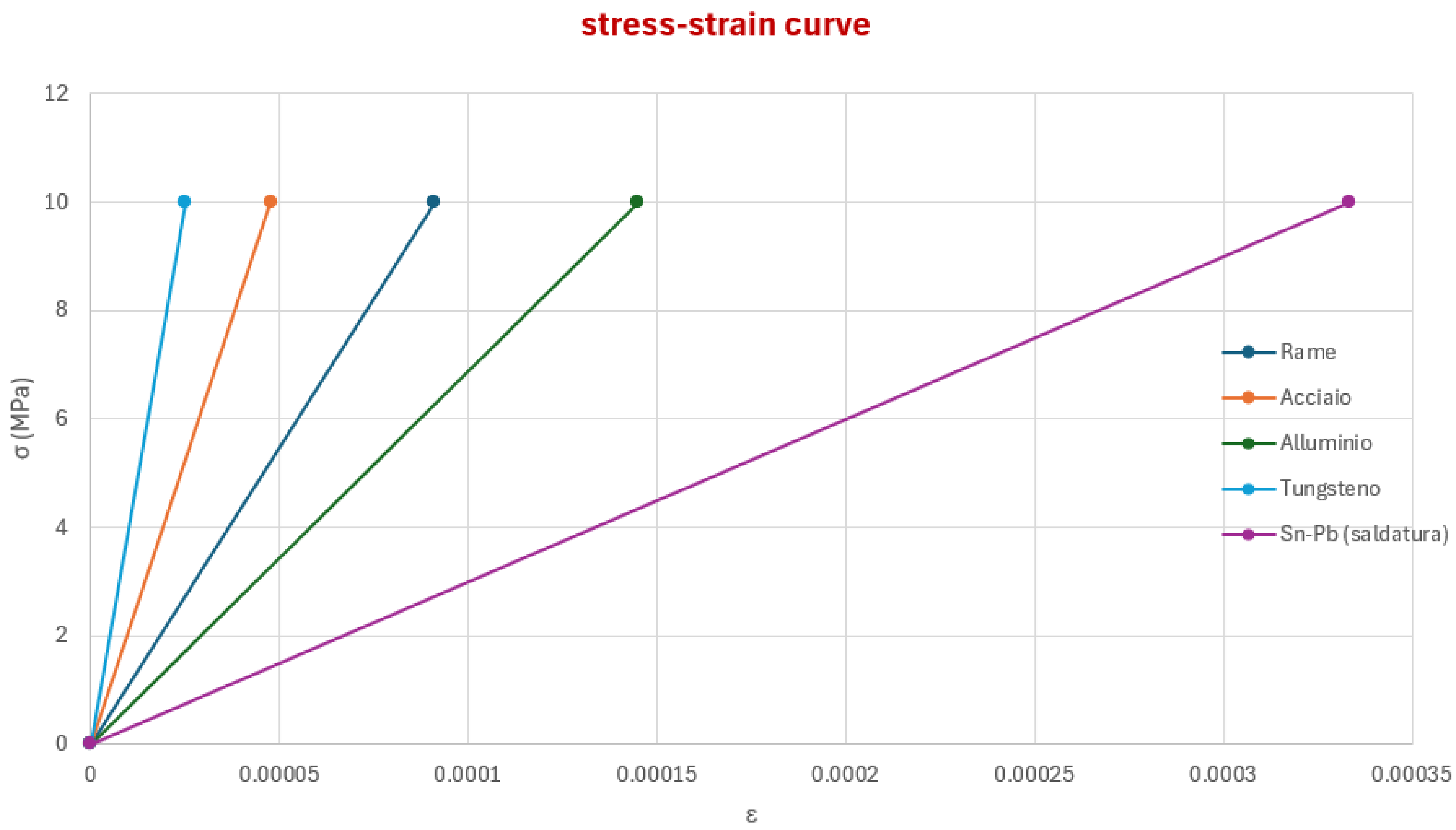
- ε strain
- ΔL total elongation [m]
- L_0 original length [m]

$$\Delta l = \varepsilon * l0$$

Dati

l0 (m)	A (mm^2)	F (N)
10	1	10

Materiale	E (Mpa)	σ (Mpa)	ε	Δl (m)
Copper	110000	10	9.09091E-05	=D8*B4
Steel	210000	10	4.7619E-05	
Aluminum	69000	10	0.000144928	
Tungsten	400000	10	0.000025	
Sn-Pb (welding)	30000	10	0.000333333	



Materiale	E (Mpa)	σ (Mpa)	ε	Δl (m)
Copper	110000	10	9.09091E-05	0.000909091
Steel	210000	10	4.7619E-05	0.00047619
Aluminum	69000	10	0.000144928	0.001449275
Tungsten	400000	10	0.000025	0.00025
Sn-Pb (welding)	30000	10	0.000333333	0.003333333

Exercise 6

Consider an aluminium cylindrical sample with a **diameter of 1cm**.
the significant values for applied force underlined in tensile test are:

- 4600 N (yield),
- 8450 N (maximum)
- 5700 N (rupture) .

When rupture occurs sample diameter is 1/3 of the initial one

Determine:

- Yield strength
- Tensile strength
- Seemly fracture strength
- True fracture strength

Initial section: $(0,5)^2 \cdot \pi \text{ cm}^2 = 0,785 \cdot 10^{-4} \text{ m}^2$.

Yield strength:
$\sigma_y = 4600/0,785 \cdot 10^{-4} = 5,86 \cdot 10^7 \text{ N/m}^2 =$ **58.6 MPa**

Tensile strength:
$\sigma_m = 8450/0,785 \cdot 10^{-4} = 10,76 \cdot 10^7 \text{ N/m}^2 =$ **107.6 MPa**

Seemly fracture strength:
$\sigma_R = 5700/0,785 \cdot 10^{-4} = 7,26 \cdot 10^7 \text{ N/m}^2 =$ **72.6 MPa**

If at the rupture sample present a diameter of 1/3 of the initial one (which is 1cm), so 0.33cm, the real section that support the load is only:

r= 0.33/2 = 0,165 cm

$(0,165)^2 \cdot \pi$ cm $^2 \Rightarrow 0,086 \cdot 10^{-4}$ m^2

So the **true fracture strength** is:

σ_R=5700/0,086 $\cdot 10^{-4}$ = 66,3 $\cdot$ 10^7 N/m^2 = **663 MPa**

If we consider the initial cross-sectional area, we obtain the *engineering stress* curve. The stress–strain curve typically shows a bell-shaped behavior.

In the *true stress* representation, the curve continues to rise due to the reduction in the cross-sectional area caused by necking.